
The Vertical Dimension of Cities

LSE GY457 - AT 2025

Topic 7

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Course overview

- **Block 1**

- Intro to Urban and Regional Economics and course overview
- Topic 1: Regional and urban concentration forces
- Topic 2: The empirics of agglomeration
- Topic 3: The costs and benefits of agglomeration

- **Block 2**

- Topic 4&5: Monocentric city model
- Topic 6: Firm location choice

- **Block 3**

- **Topic 7: The vertical dimension of cities**
- Topic 8: Neighborhoods and segregation
- Topic 9: Gentrification and suburbanization
- Topic 10: Measuring cities with new technologies

Recap from last time

- Firm bid-rent can be derived using zero-profit condition
 - Historically, high transportation costs relevant
 - Agglomeration spillovers more plausible determinant today
- Changes in transport technology led to employment decentralization
 - Manufacturing employment decentralizes due to lower transportation costs
 - Services can decentralize since agglomeration is endogenous
- Decentralized vs. dispersed employment
 - Subcenters can emerge endogenously if cities get larger
 - Developers strategically build edge cities to make profits

Today: The vertical dimension of cities

- Model
 - Vertical and horizontal costs and benefits
 - General equilibrium
- Do rents increase in height within buildings?
 - Returns to height
 - Costs of height
- Equilibrium and counterfactuals
 - Urbanization as a cause of skyscraper development
 - Skyscraper development as cause of urbanization
 - Height regulation

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London's tallest building until 1963...



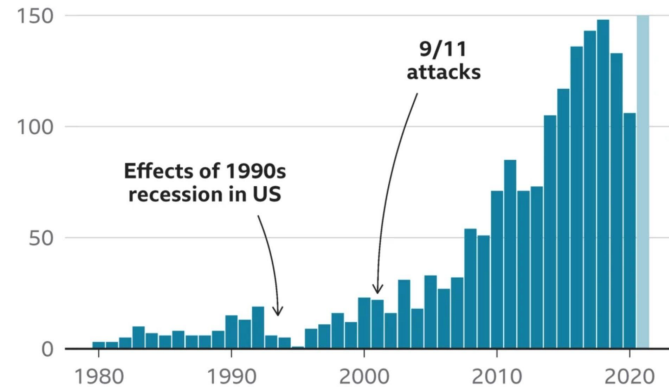
- Tallest building in London for more than two centuries
 - Between 1708-1963
- 111 meters tall
- Today ranked 53rd
 - The Shard (310 m) dominates skyline since 2012

Aerial photo of St. Paul's Cathedral (Source: St. Paul's Cathedral website/David Morse)

Rise of tall buildings in past 50 years

The rise of the skyscraper 🏙️

Number of buildings over 200 metres high completed globally



Note: The figure for 2021 is projected to be between 125 and 150 buildings

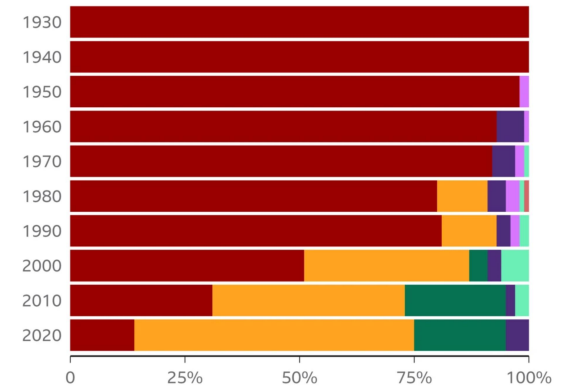
Source: Council on Tall Buildings and Urban Habitat

BBC

Asia now has most of the tallest skyscrapers

% share of the world's 100 tallest buildings, by continent

North America Middle East South America Africa
Asia Europe Oceania



Source: Council on Tall Buildings and Urban Habitat

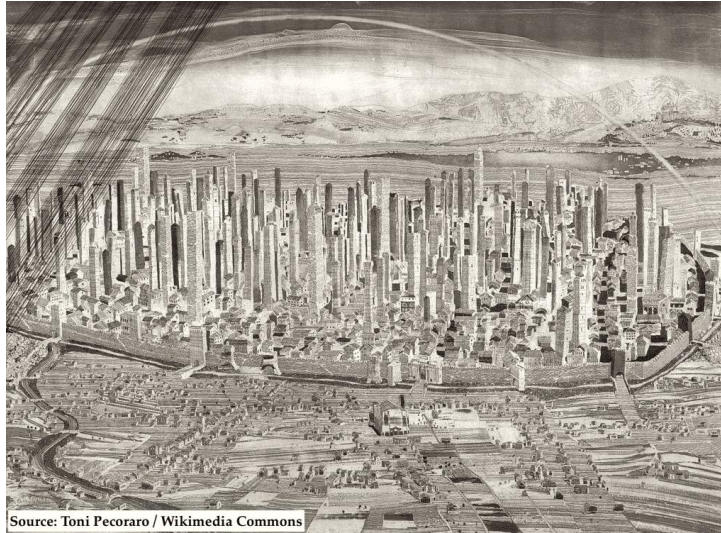
BBC

Source: [Does the skyscraper still have a future?](#) (BBC, 4 November 2021)

How to incorporate building heights into a urban model

- Ingredients:
- Endogenous land use (endogenous CBD has a mass)
 - Need residents and firms location choices
- Endogenous population (open city framework)
 - Takes into account costs and benefits of urbanization
- Endogenous wages (respond to agglomeration forces)
- Account for costs and benefits of building heights
 - NEW!
 - Can differ by land use
- General equilibrium

Are building height decisions rational?



- Bologna “la Turrita” in Middle Ages probably had around 100 towers
- Main tower: Asinelli, 97.2 m tall, built in 1109-1119, 1.3° off-center lean
- vs. Garisenda tower: today 48 m (was 60, “collapsed” in 1300s), 4° lean

Height competition

- Helsley and Strange (2008)
 - There is intrinsic value of being the tallest building
 - Analysis using game theory shows that skyscrapers are developed beyond the fundamentally efficient height
 - Developers build excessively tall to discourage competition
 - Empire State Building has been the tallest building for ~40 years
- Evidence is mixed
 - Some studies seem to suggest that building heights generally follow profit maximizing motives (Barr, 2010)
 - Others that is result of height competition (Barr, 2012) or skyscraper competition between cities (Barr, 2013)

Are skyscrapers too tall? (Ahlfeldt-Barr, 2022)

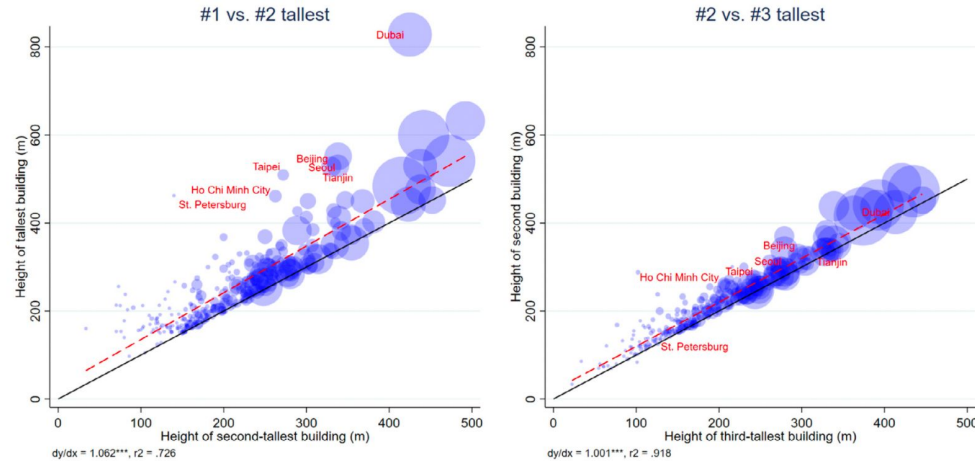


Fig. 4. Tallest building heights Note: Data covers 315 cities around the world with at least one skyscraper (heights ≥ 150 m. Marker size is proportionate to the number of skyscrapers in a city. Data are from Emporis (accessed in April, 2021).

- Tallest buildings often claimed to be irrationally high
 - But only a few skyscrapers seem to dominate cities skyline
 - Within cities, 2nd & 3rd tallest building have almost same height

Are skyscrapers too tall? (Ahlfeldt-Barr, 2022)

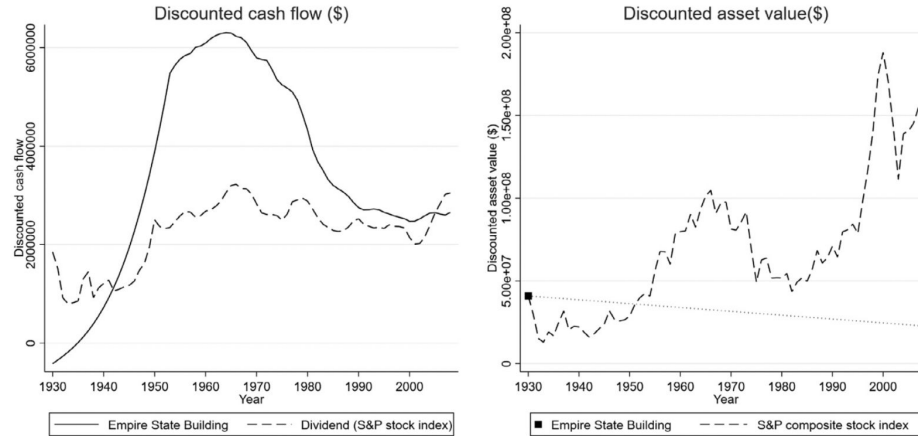


Fig. 5. Empire State Building vs. stock market Note: ESB asset value is total cost of structure and land in 1930 and land value in 2008 (as we assume that the structure has fully depreciated). For stock market, the asset value is the total investment inflated by the stock market index. Cash flow of the ESB is the net operating income (EBITDA). Cash flow of the stock market is the dividend (the product of the dividend yield and the asset value). All time series deflated by the cumulated opportunity cost of capital (risk-free rate). The net realized return for the ESB, at 5.4%, beats the stock market, at 4.3%. See Online Appendix Section B.4 for details.

- Empire State Building claimed to be irrational outcome of 1920s *skyscraper race* (but performed well after GD)

Model: Geography (Ahlfeldt and Barr, 2022)

- Adopt stylized geography as monocentric city model
- Location is defined by x (distance to the historic center)
 - $x = 0$ in the historic center
 - $D(x) = x$ is distance from the historic center
- Production amenity declines with distance from center
 - Capture agglomeration spillovers
- Residential amenity declines with distance from center
 - Why? Can think of it as capturing commuting costs

Model: Workers (Ahlfeldt and Barr, 2022)

- Workers utility depends on floor space (f_R), other goods (g), and residential amenities (A_R)
- $U(x,s) = A_R(x,s) g^\alpha f_R(x,s)^{1-\alpha}$
 - Utility depends on horizontal (x) and vertical (s) space
 - Assumption 1: A_R increases in s and decreases in x
 - Assumption 2: consumption of f_R depends on s,x (how?)
- Budget constraint: $y = g + p_f(x,s) f_R(x,s)$
 - Where we normalize the price of basket of goods g to 1 (i.e., $p_g = 1$)
- Horizontal bid-rent declines in x and depends on s
- *How does this setup differ from the canonical monocentric city model?*

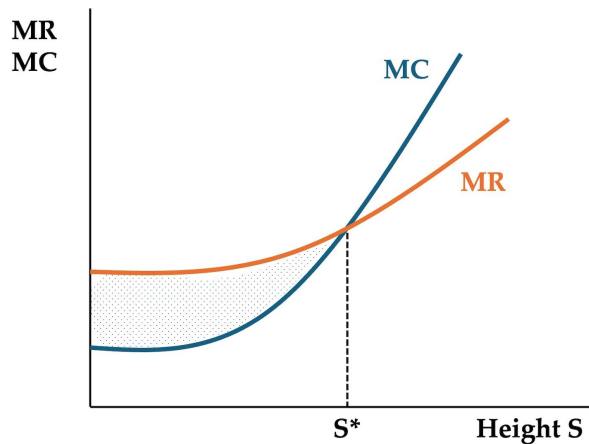
Model: Firms (Ahlfeldt and Barr, 2022)

- Assume Cobb-Douglas production function:
$$g(x,s) = A_C(x,s) L^\beta f_C(x,s)^\beta$$
 - Tradable goods g produced using labor L and commercial floor space f_C as inputs
 - Assumption 1: Production amenity A_C increases in s and decreases in x (*why? what is the intuition?*)
 - Assumption 2: consumption of floor space f_C depends on s, x
 - Symmetric to worker utility maximization problem
- Profit maximization: $\pi(x,s) = g(x,s) - yL - p_C(x,s)f_C(x,s)$
- Firms horizontal bid-rent decreases in x , depends on s

Model: Developers (Ahlfeldt and Barr, 2022)

- Choose optimal buildings height S to maximize profits
 - Developers profits: $\pi_U(x, S(x)) = p(x)S(x) - c(S(x))S(x) - r(x)$
 - $p(.)$ is horizontal bid rent (determined in equilibrium) = $p_U S(x)^\omega$:
 p_U is rent of a one-story building; ω is height elasticity of rent
 - S is buildings height
 - $c(.)$ is construction cost = $c_U S(x)^\theta$: c_U is unit cost of a one-story building; θ is height elasticity of construction costs
 - $r(.)$ is land rent function
 - Assumption 1: c is convex in building height S (*why?*)
 - Assumption 2: At any location x , there can be only one building, either for commercial or residential use
 - Optimal building height $S^*(x)$ decreases in height construction c and increases in A_R, A_C (*intuition?*)
-

Model: Land rents (Ahlfeldt and Barr, 2022)



- Developers are in perfectly competitive market
- In equilibrium, zero profits
- Profit maximizing height S^* is when marginal revenues (MR) equals marginal costs (MC)
 - $\partial\pi/\partial S = 0$
- In the dotted area, profits per plot of land end up being fully capitalized into land rent
 - Since developers compete (bid) for land

Model: Equilibrium (Ahlfeldt and Barr, 2022)

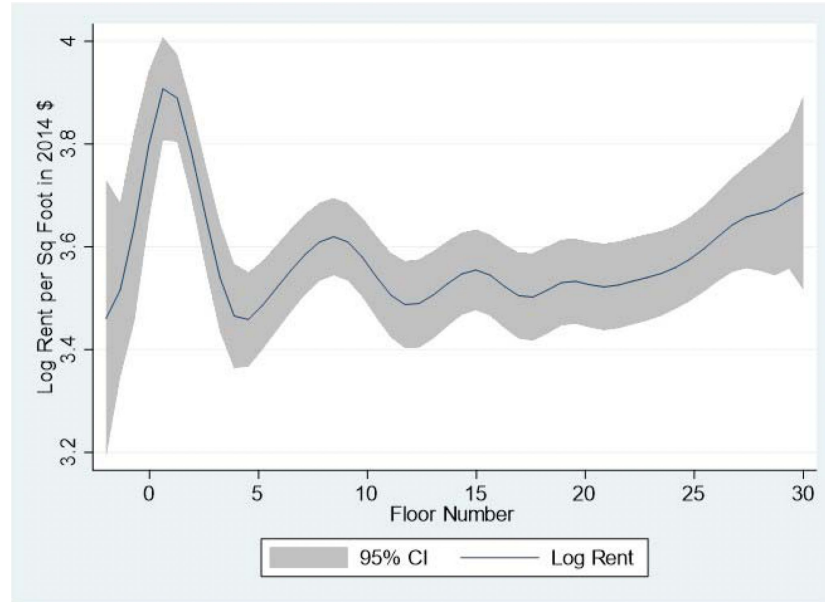
- In each location, land use is allocated to the highest bidder
 - Commercial vs. residential zones (land use pattern)
- Real estate and labor markets clear
 - What does that mean?
 - 1. Total floor space supplied by developers is occupied either by firms or by workers (i.e., supply equals demand of floor space)
 - 2. Total workers employed by firms must equal total number of workers living in the city
- Equilibrium consists of solving for workers, residents, floor space rent, land rent, building height in each location + city-level wage and total population (GE!)

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Commercial vertical rent gradient

(Liu et al., 2018)



- What does the figure suggest us?

Commercial vertical rent gradient

(Liu et al., 2018)

- Liu et al. (2018) estimate elasticity of rent wrt floor
 - Different from elasticity of average rent wrt building height
- Average rent of building with floors $s=1 \dots S$: $p_s = (\bar{p}/S) \sum_s s^\alpha$
 - $\bar{p}$ is a common price component (independent of floor level)
 - α is elasticity of rent wrt floor level
 - Compute for a series of buildings with $S = 1$ to $S = 50$ floors
- Compute the elasticity of rent wrt to height elasticity
 - $\ln(p_s) = c + \omega \ln S + \varepsilon$
- α : How does rent vary with floor level (*within building*)?
- vs. ω : How does rent vary with height (*across buildings*)?

Commercial vertical rent gradient

(Liu et al., 2018)

	Offering Memo Data		CompStak Data	
	Double Log	Semi-Log	Double Log	Semi-Log
Below ground floor	-0.2621 (0.1035)	-0.4596 (.0987)	- -	- -
Ground floor	0.4661 (0.0952)	0.3448 (0.1002)	0.1156 (0.0357)	0.0295 (0.0354)
Ground Floor X Bldg Height	0.0046 (0.0017)	0.0059 (0.0018)	0.0070 (0.0017)	0.0073 (0.0017)
Log(Floor number + k) ^b	0.1883 (0.0355)	- -	0.0858 (0.0049)	- -
Floor number	- -	0.0087 (0.0012)	- -	0.0058 (0.0003)
Observations	5,445	5,445	37,007	37,007
Lease quarter Fixed Effects	-	-	Yes	Yes
Building Fixed Effects	93	93	1,922	1,922
R-sq within	0.162	0.177	0.247	0.254

^a Dependent variable for the OM regressions is gross rent per square foot in \$2014. Dependent variable for the CS regressions is in \$2014 and is net rent which adjusts gross rent for months of free rent at the start of the lease and other accommodations. Standard errors clustered at the building level are in parentheses.

^b k is set to a value 1 unit larger in absolute value than the lowest basement floor in the data, -5 for the offering memo data and -1 for the CompStak data.

- Rent gradients with building fixed effects
- Using CS data:
 $\alpha = 8.58\%$
- Ground floor premium up to around 13%
($=e^{0.1156} - 1$)
- Using this data,
 $\omega = 7.1\%$

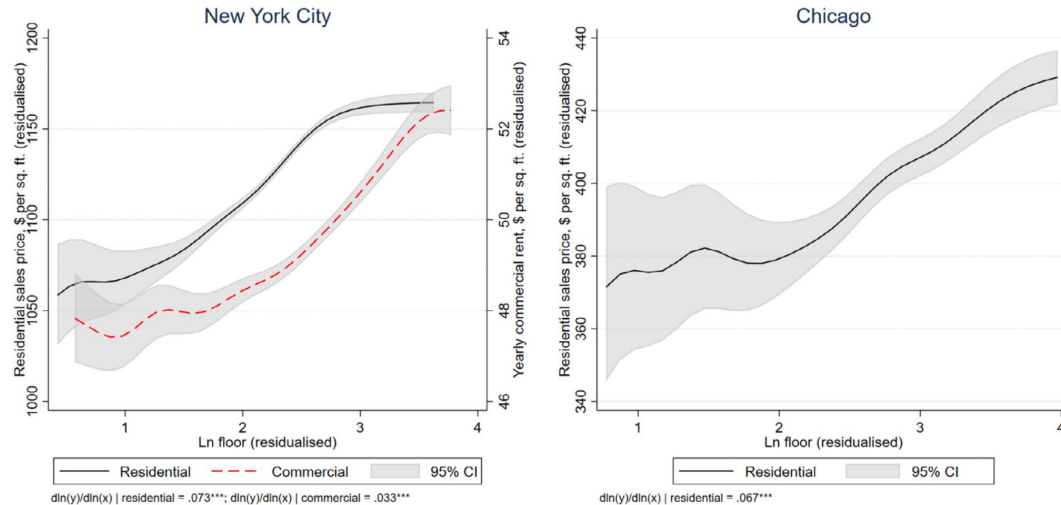
Commercial vertical rent gradient

(Liu et al., 2018)

	Offering Memo Data			CompStak Data		
	(1)	(2)	(3)	(4)	(5)	(6)
	Floors 3 through 29	Floors 30 through 59	Floors 60 and above	Floors 3 through 29	Floors 30 through 59	Floors 60 and above
PANEL A: Double Log	▶			▶		
Log(Floor number + k) ^b	0.1334 (0.0322)	0.1810 (0.1571)	1.036 (0.0196)	0.0760 (0.0046)	0.2873 (0.0394)	1.274 (0.3036)
Observations	3,524	774	115	29,360	4,602	116
Lease quarter Fixed Effects	No	No	No	Yes	Yes	Yes
Building Fixed Effects	93	44	4	1,862	369	18
R-sq within	0.029	0.008	0.112	0.257	0.295	0.710

- Elasticity of office rent wrt floor level is higher at higher floors
- Highly convex rent premium

Residential vertical rent gradient (Ahlfeldt and Barr, 2022)



Positive gradient due to higher utility (nicer view, less noise)

Residential vertical rent gradient (Ahlfeldt and Barr, 2022)

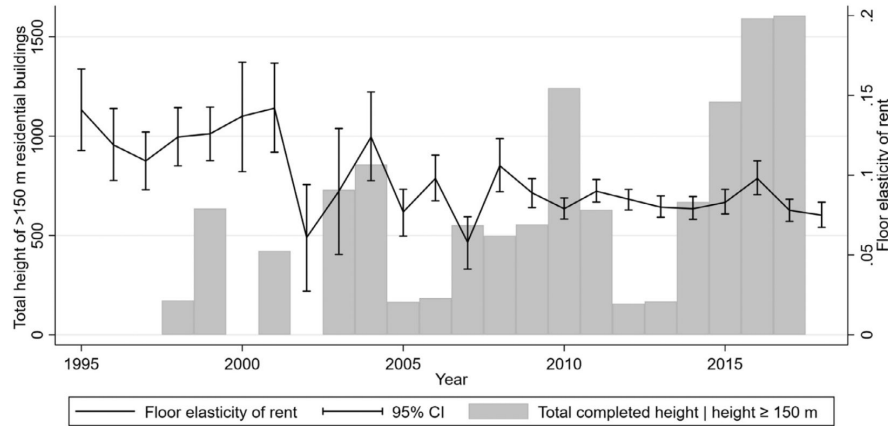


Fig. 9. Returns to height over time in the New York City residential market Note: The floor elasticity of rent is estimated in regressions of log rent against log floor by year, controlling for unit characteristics and building fixed effects. Confidence bands are at the 95% level. Residential rents are implied rents based on apartment prices data from StreetEasy, assuming a constant discount rate. Total completed height of residential buildings exceed 150 m are from the CTBUH Skyscraper Center.

Residential vertical rent gradient decreased over time. *Why? Do you think the same pattern applies to cities other than NY?*

Residential vertical rent gradient: London (2022 REEF dissertation)

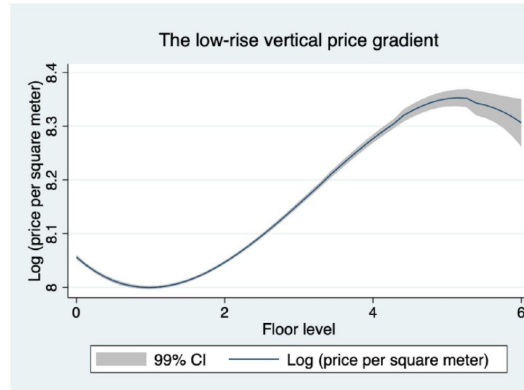


Figure 2: The non-linear vertical price gradient was yielded from estimating locally weighted regressions using Stata's `lpolyci` function. Stata's default Epanechnikov Kernel function is used in the analysis. Inspired by [Liu et al. \(2018\)](#), the bandwidth of 1.5 and function degree of three are chosen. Confidence bands are at the 99% level. Data source: [\(Dennett, et al., 2021\)](#)

$\omega = 0.6\%$

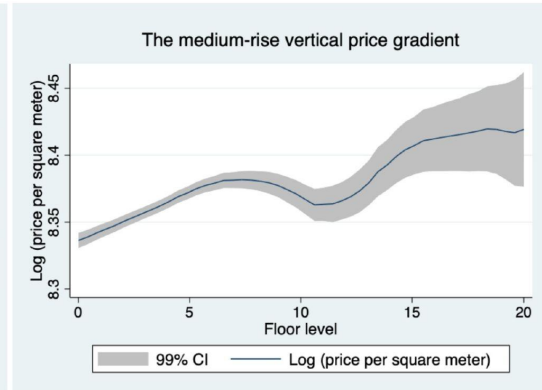


Figure 1: Stata's default polynomial smooth of degree 0 and the Epanechnikov Kernel function are used to yield the mid-rise VPG, with a bandwidth of 1.5.

Confidence bands are at the 99% level.

Data source: [\(Dennett, et al., 2021\)](#)

$\omega = 3\%$

Source: "The vertical price gradient in London" (Candidate number 40037)

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Are taller buildings more expensive to build?

- Liu et al. (2018) provide estimates of the vertical rent gradient conditional on building fixed effects
 - Control for unobserved building characteristics (location, quality, design, maintenance cost, sustainability, etc.)
 - Gives credibility to their estimates (causal identification)
- For residential buildings, Himbert and Danton (2018):
 - Fifth floor exhibit 5% premium over first floor
 - Ground floor premium equals to 1.5-3.5% (*why?*)
- Additional evidence (without building fixed effects) reinforces the idea of height premium
 - Colwell et al. (1998), Koster et al. (2014), Shilton-Zaccaria (1994)

Height elasticity of construction cost

(Ahlfeldt and McMillen, 2018)

- Rule-of-thumb: Construction costs per unit of floor space increase by 2% for each additional floor
 - London Department of the Environment (1971)
- Ahlfeldt and McMillen (2018): *Is this estimate confirmed in the data?*
 - First paper to estimate elasticity of construction cost using data
- Regression: $\ln(K/F)_{it} = \alpha_t + \theta \ln(S_{it}) + \varepsilon_{it}$
 - Dataset is from Emporis, where i is building, t is time (year)
 - K is total construction cost (excluding land acquisition cost)
 - F is total building floor space
 - S is measure of building height

Height elasticity of construction cost (Ahlfeldt and McMillen, 2018)

TABLE 5.—CONSTRUCTION COST ELASTICITY ESTIMATES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Log Floor Space Cost							
Log building height	0.251*** (0.006)	0.475*** (0.032)	0.634*** (0.040)	0.541*** (0.034)	0.612*** (0.034)	0.523*** (0.036)	0.611*** (0.041)	1.729*** (0.050)
Country effects	Yes	Yes	Yes	-	-	-	-	-
Decade effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	-
Mean number of floors	2.5	15.9	15.6	12.3	9.2	20.600	20.100	110
Semielasticity	0.099	0.030	0.041	0.044	0.067	0.025	0.030	0.016
Data	Obs.	Obs.	Obs.	Obs.	Obs.	Obs.	Obs.	Engin.
Building use	All	Com.	Res.	Com.	Res.	Com.	Res.	Com.
Building type	Small	Tall	Tall	Tall	Tall	Tall	Tall	ST
Region	World	World	World	US	US	PC	PC	-
R^2	0.109	0.492	0.614	0.620	0.451	0.730	0.522	0.997
N	32,016	1,219	1,570	818	1,402	818	1,401	9

Observed data (obs.) are from Emporis. Engineering estimates (Engin.) are from Lee et al. (2011). Ln Floor space cost is computed using predicted usable floor space values based on the regressions reported in appendix section 4.8. Small buildings have fewer than five floors. Tall buildings have five or more floors. Super-tall buildings (ST) have ninety or more floors. The semielasticity is computed by dividing the cost elasticity by the mean number of floors. Pseudo-Chicago (PC) sample is the U.S. sample, reweighted to resemble the distribution of ln building heights in Chicago (using a propensity score matching; see appendix section 4.9). For a small percentage of buildings, height is imputed based on floors using an auxiliary regression of height against floors (on average height increases by 3.6 meters per floor). Standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

- Height elasticity of construction cost increases in building height ($\theta > 0$)
- 2% rule of thumb roughly applies to tallest buildings (semi-elasticity)
- Cost of height greater for residential than commercial buildings (why?)

Height elasticity of construction cost (Ahlfeldt and Barr, 2022)

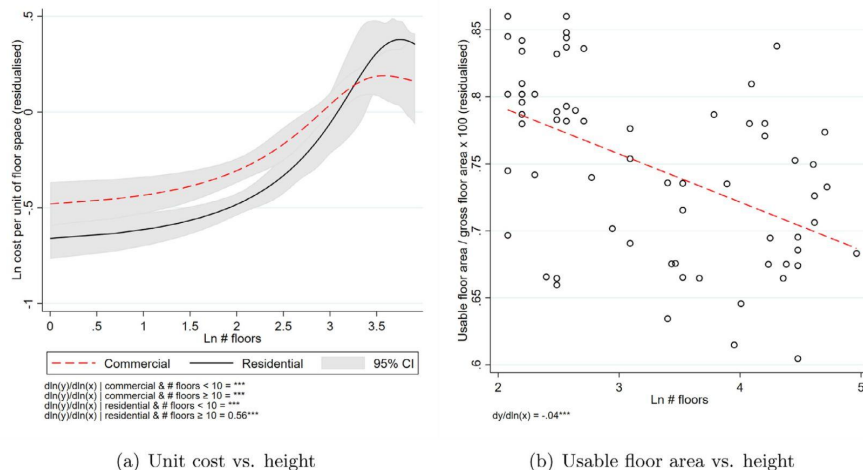


Fig. 6. Cost of height Note: In panel (a), we first regress the log of the ratio of building construction cost over building floor space against decade fixed effects, country fixed effects, and number-of-floors fixed effects. The displayed non-linear functions are the outcome of locally weighted regressions of the estimated number-of-floor fixed effects against the number of floors, using a Gaussian kernel and a bandwidth of 10. Confidence bands are at the 95% level. The vertical line marks the natural log of 10. Data are from <https://www.emporis.com/> (see Ahlfeldt and McMillen (2018) for details). In panel (b), each marker is an individual building. The dependent variable is residualized in regressions against country fixed effects and a time trend. Note that the natural log of 10 (floors) is about 2.3. Observations were culled from Sev and Özgen (2009), Watts et al. (2007), Kim (2004), and Berger (1967).

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Urban gradients (Ahlfeldt and Barr, 2022)

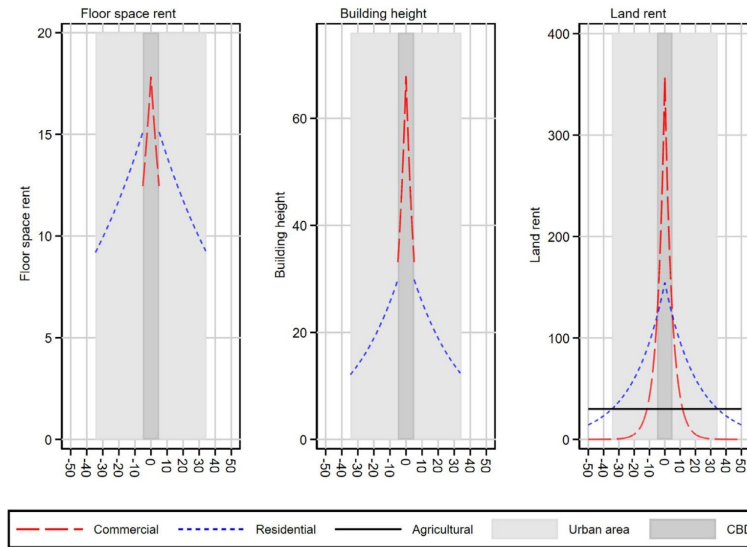


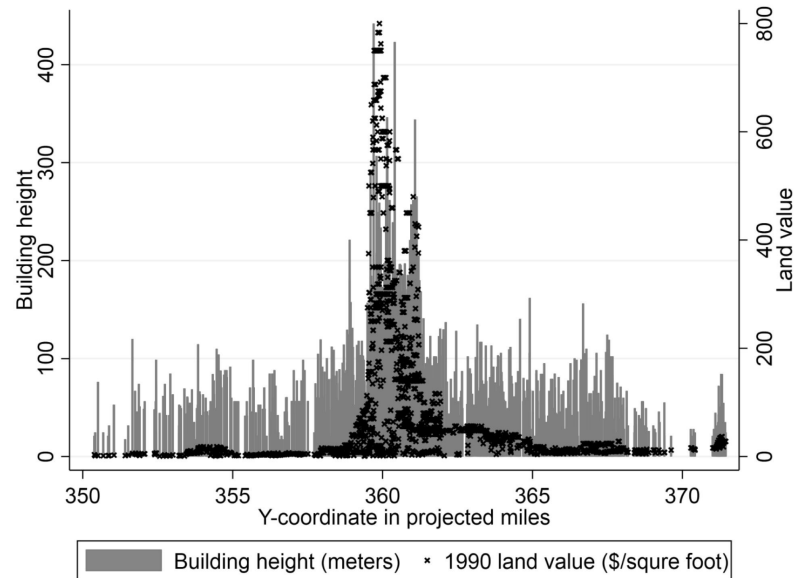
Fig. 10. Urban gradients Note: The figure illustrates the solution to the model laid out in Section 3 using the parameter values reported in Table 1. Floor space rent is the average per-unit rent within a building $\bar{p}^U(x)$. The x-axis gives the distance from the historic center in units that roughly correspond to kilometers in real-world cities.

Equilibrium city structure

(Ahlfeldt and Barr, 2022)

- Relative floor space rent offsets locational advantages
 - Decreases in distance to CBD due to commuting costs and decay of agglomeration spillovers
- Building height adjusts to floor space rents
 - Different optimal heights for different land uses
- Land rents determined by floor space rent and height
 - Different land rents for different uses
- Land use allocated to highest bidder
 - Determines size of CBD and residential zone
- Level of floor space rent adjusts to clear markets
 - So that aggregate demand equals supply, by use

Land price and building height in Chicago (Ahlfeldt and McMillen, 2018)



Today: The vertical dimension of cities

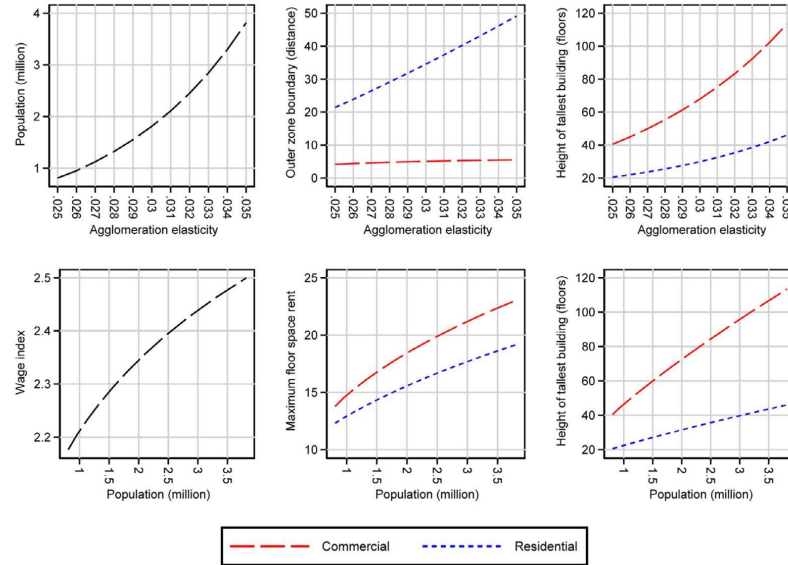
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Comparative statics: Urbanization

(Ahlfeldt and Barr, 2022)

- Are skyscrapers a cause or an effect of urbanization?
- To understand the causal effect of urbanization
 - Solve model for varying values of agglomeration elasticity (i.e., we change the degree of urbanization)
 - At baseline, $\beta=0.03$ (from literature); consider $\beta \pm 0.05$
 - Returns to living in cities increase (industrialization, transition to knowledge-based tradable service economy)

Comparative statics: Urbanization (Ahlfeldt and Barr, 2022)



- Wages $\uparrow$
- Pop $\uparrow$
- City size $\uparrow$
- Rent $\uparrow$
- Height $\uparrow$

Urbanization
causes vertical
(and horizontal)
growth

Fig. 13. Variation in external returns to agglomeration Note: We report solutions to the model developed in Section 3 under varying values of the agglomeration elasticity, β . All other parameter values are kept constant at the levels reported in Table 1.

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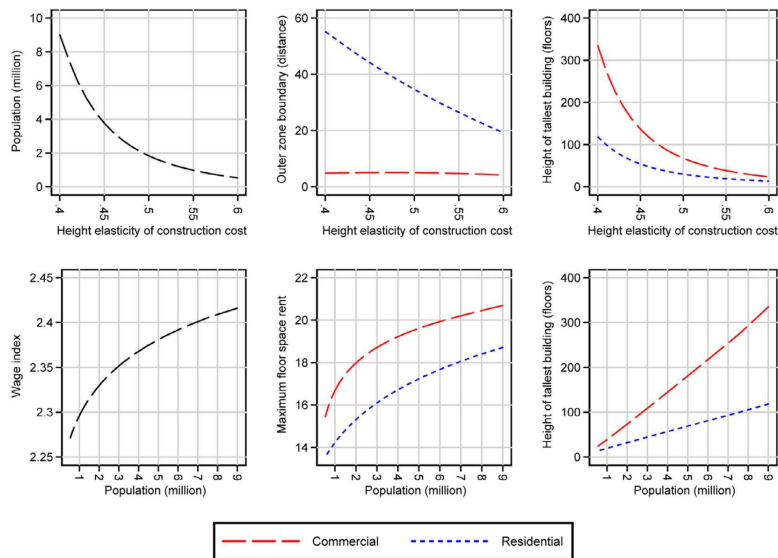
Comparative statics: *Skyscraperization*

(Ahlfeldt and Barr, 2022)

- Are skyscrapers a cause or an effect of urbanization?
- To understand the causal effect of tall buildings
 - Solve model for varying values of height elasticity of construction cost (i.e., we change the degree of *skyscraperization*)
 - At baseline, $\theta^C = 0.5$ & $\theta^R = 0.55$ (from literature); consider $\theta^C \pm 0.1$ and adjust θ^R to keep relative distance constant (i.e., $\theta^C - \theta^R = 0.05$)
 - Cost of height decreases (innovation in construction technology, elevator, steel frame, mainframe computing,...)

Comparative statics: *Skyscraperization*

(Ahlfeldt and Barr, 2022)



- Height ↑
- Rent ↓
- Pop ↑
- City size ↑
- Wages ↑
- Rent ↑↑

Vertical growth
causes urbanization

*Extra: What would
happen in GE if
transport costs↑?*

Fig. 15. Variation in cost of height Note: We report solutions to the model developed in Section 3 under varying values of the amenity decay parameters $\{\theta^C, \theta^R\}$. We keep the relative size constant at $\theta^C - \theta^R = 0.05$. All other parameter values are kept constant at the levels reported in Table 1.

Urbanization vs. *Skyscraperization*

(Ahlfeldt and Barr, 2022)

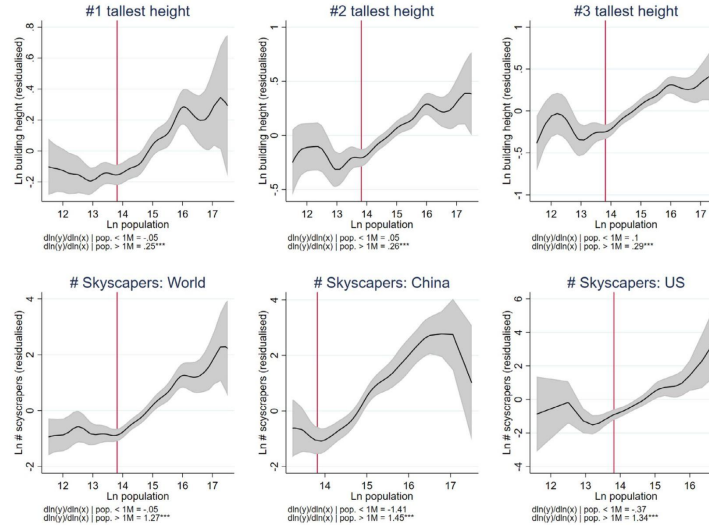


Fig. 16. Skyscrapers and city size Note: Skyscrapers and population for cities around with world, with at least one 150 m+ skyscraper. $\ln(\text{building height})$ and $\ln(\text{\# skyscrapers})$ are residualized in regressions against country fixed effects. The black solid lines are from locally weighted regressions using a Gaussian kernel and a bandwidth of 0.25. Confidence bands are at the 95% level. Vertical line marks the log of one million. Skyscraper data is from <https://www.skyscrapercenter.com/>; accessed Feb. 2020. Population data are from several sources (available upon request) and are the most currently available counts for the metropolitan regions, which includes the central city and surrounding population agglomerations.

- From previous results, we expect positive relation between pop and vertical city-size
- Indeed found in the data (figure)
- Originates from demand side (urbanization) and supply side (skyscraperization)

Today: The vertical dimension of cities

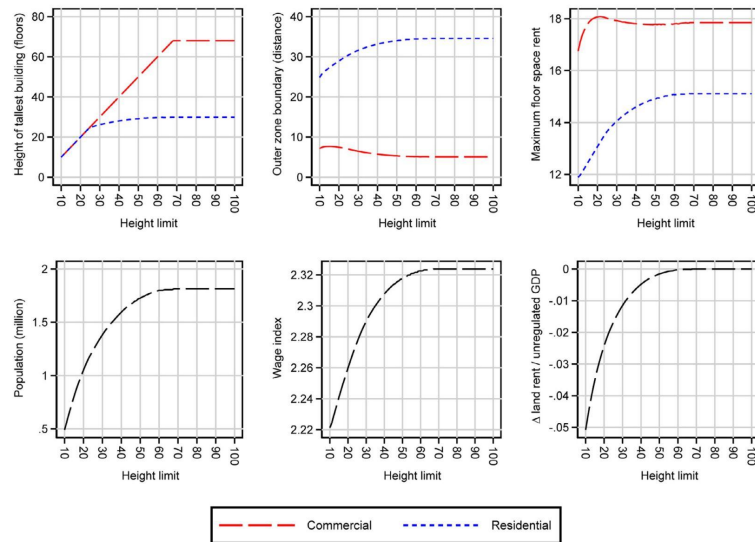
- Model
 - Vertical and horizontal costs and benefits
 - General equilibrium
- Do rents increase in height within buildings?
 - Returns to height
 - Costs of height
- Equilibrium and counterfactuals
 - Urbanization as a cause of skyscraper development
 - Skyscraper development as cause of urbanization
 - Height regulation

Comparative statics: Height regulation

(Ahlfeldt and Barr, 2022)

- What happens if we introduce height regulations?
- Model it as imposing a binding height limit
 - Optimal building height that developers choose: $S^{**} = \min(S^*, \bar{S})$
 - Where S^* is profit maximizing height (as we saw before)
 - $\bar{S}$ is the height limit imposed by the regulation
- To simulate effect of height limits in open-city model
 - Solve model for varying degrees of regulation $\bar{S} = (10, 100)$
 - Compute welfare effects
 - Welfare effects measured as reduced land rent (“landowner revenue”) since land is immobile vs. population adjusts by leaving the city in response to lower utility

Comparative statics: Height regulation (Ahlfeldt and Barr, 2022)



- Height ↓
- Rent ↑
- Pop ↓
- City size ↓
- Wages ↓
- Rent ↓↓
- Welfare ↓

Fig. 17. Variation in height limit Note: We report solutions to the model developed in Section 3 under varying values of the height limit, $S^R = S^C$. All other parameter values are kept constant at the levels reported in Table 1. In the bottom-left corner, we measure aggregate land rent over the area occupied by the unregulated city. We do not vary this interval across simulation runs and aggregate over the maximum land bid rent, $\max(r^C(x), r^R(x), r^U(x))$, at each location. GDP is the wage bill, $(y^U N)$, weighted by the inverse of the labor factor share, α^C .

Regulations: Developers vs. planners

- Many cities impose binding height regulations
 - Historically to protect views, control density, fire protection,...
- Tension between developers and planners
 - Developers may want to build taller to maximize profits
 - Planners may be concerned with negative externalities
- How can developers “convince” planners?
 - Developers can make some form of “social contribution”
 - e.g., social housing, open spaces, aesthetics, ...others?
 - Or: lobbying/corruption

The role of “Trophy architects”

- Profit maximizing building height in City of London: estimated at 156 floors
 - In tendency, new office building: only 8 floors
- To convince planners, one can hire trophy architects
 - Architects winning most prestigious awards
 - e.g. to address planners' concerns about negative visual effects
 - In London: Renzo Piano and the Shard?
- Empirical findings:
 - With TA, planners on average approve 14 additional floors
 - Large source of rental revenues!
 - But trophy architect adds ~13% of building costs (*more in seminar*)

Summary

- Vertical rent gradient
 - There is a ground floor premium driven by retail
 - Positive price gradient at higher floors driven by offices
- Can height elasticities be used to find profit-maximizing building height?
 - Marginal average unit cost equals marginal average unit rent
 - Determine maximum bid for a given internal rate of return
- Spatial equilibrium
 - Skyscrapers are both a cause and an effect of urbanization
 - Height regulations lead to welfare loss, but not necessarily higher rents

Next week: Nhbds and segregation

- Housing and neighborhood characteristics
 - Urban housing and hedonic models
 - Impact of neighborhood characteristics on property prices
 - Housing demand and discrimination in lending market
- Theories of segregation
 - Rise and decline of the American ghetto
 - Causes of segregation (Boustan, 2010)
 - Race-based tipping dynamics
- Impacts of transportation infrastructure on segregation
- Consequences of segregation and its implications
 - Role of ethnic enclaves

Thank you!

Readings and references

Required readings

- Ahlfeldt, G. M., and Barr, J. (2022). The economics of skyscrapers. A synthesis. *Journal of Urban Economics*, 129, 103419..
- Ahlfeldt, G. M., and McMillen, D. P. (2018). Tall Buildings and Land Values: Height and Construction Cost Elasticities in Chicago, 1870 –2010. *Review of Economics and Statistics*, 100(5), 861–875.
- Liu, C. H., Rosenthal, S. S. and Strange, W. C. (2018). The vertical city: Rent gradients, spatial structure, and agglomeration economies. *Journal of Urban Economics*, 106, 101–122.

Other readings and references

- Barr, J.. (2010). Skyscrapers and the Skyline: Manhattan, 1895–2004. *Real Estate Economics*, 38(3), 567-597.
- Barr, J. (2012). Skyscraper Height. *The Journal of Real Estate Finance and Economics*, 45(3), 723-753.
- Barr, J.. (2013). Skyscrapers and skylines: New York and Chicago, 1885-2007. *Journal of Regional Science*, 53(3), 369-391.
- Colwell, P. F., Munneke, H. J., and Trefzger, J. W. (1998). Chicago's Office Market: Price Indices, Location and Time. *Real Estate Economics*, 26(1), 83-106.
- Danton, J., and Himbert, A. (2018): Residential Vertical Rent Curves, *Journal of Urban Economics*.

Readings and references

- Department of the Environment (1971), Local Authority Offices: Areas and Costs. London: DOE, 1971.
- Helsley, R. W. and Strange, W. C. (2008). A game-theoretic analysis of skyscrapers. *Journal of Urban Economics*, 64(1), 49-64..
- Koster, H. R. A., van Ommeren, J., and Rietveld, P. (2014). Is the sky the limit? High-rise buildings and office rents. *Journal of Economic Geography*, 14(1), 125-153.
- Shilton, L., and Zaccaria, A. (1994). The avenue effect, landmark externalities, and cubic transformation: Manhattan office valuation. *The Journal of Real Estate Finance and Economics*, 8(2), 151-165.